AMITY UNIVERSITY ONLINE, NOIDA, UTTAR PRADESH

In partial fulfillment of the requirement for the award of degree of
Master of Business Administration (IT)

TITLE:
AI-Based Customer Churn Prediction and Retention Strategy System

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DECLARATION

I, Nakshbh Tak, a student pursuing Master of Business Administration (IT) at Amity University Online, hereby declare that the project work entitled "AI-Based Customer Churn Prediction and Retention Strategy System" has been prepared by me during the academic year 2025-2026 under the guidance of Mr. Vikas Kumar.

I assert that this project is a piece of original bona-fide work done by me. It is the outcome of my own effort and that it has not been submitted to any other university for the award of any degree.

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CHAPTER 1: INTRODUCTION TO THE TOPIC

1.1 Background

In today's highly competitive business environment, customer retention has emerged as one of the most critical challenges faced by organizations across industries. Customer churn, defined as the phenomenon where customers discontinue their relationship with a business, leads to significant revenue losses and increased customer acquisition costs. Studies suggest that acquiring a new customer costs five to seven times more than retaining an existing one, making churn prediction a high-priority business problem.

Artificial Intelligence (AI) and Machine Learning (ML) have revolutionized the way businesses analyze data and make decisions. AI-based predictive models have become powerful tools for identifying at-risk customers before they actually leave by analyzing historical patterns, behavioral signals, and transactional data to predict the probability of churn.

1.2 What is Customer Churn?

Customer churn refers to the rate at which customers stop doing business with a company over a given period. It can be categorized as:

- Voluntary Churn: When a customer actively chooses to leave the service.
- Involuntary Churn: When a customer is forced to leave due to payment failures or service termination.
- Hidden Churn: When a customer continues to hold an account but stops actively using the service.

1.3 Role of AI in Churn Prediction

Machine learning algorithms such as Logistic Regression, Random Forest, and XGBoost can learn complex patterns from large datasets simultaneously processing dozens of variables — from customer tenure and payment history to service usage patterns and contract types — and output a churn probability score for each customer.

1.4 Importance of the Topic

The telecom industry serves as an ideal case study for churn prediction. Average churn rates in the telecom sector range from 15% to 25% annually. Even a 1% reduction in churn can result in millions of dollars in saved revenue. This project builds an end-to-end AI-based churn prediction system using the Telco Customer Churn dataset with real model results achieving 80.48% accuracy.

1.5 Justification for Topic Selection

- High Business Relevance: Customer churn is a universal problem affecting businesses across telecom, banking, insurance, and retail sectors.
- Growing Demand for AI Skills: Expertise in predictive modeling is among the most in-demand skills in the job market today.
- Real-world Dataset: The Telco Customer Churn dataset from Kaggle provides authentic structured data for realistic model building.
- End-to-End Learning: This project covers the complete data science pipeline from data collection to model deployment.
- Economic Impact: Reducing churn benefits businesses through higher revenue and customers through better service quality.

CHAPTER 2: REVIEW OF LITERATURE

2.1 Introduction

Customer churn prediction has been an active area of research since the early 2000s. Researchers across multiple domains have explored various machine learning and statistical techniques to predict churn with high accuracy. This chapter reviews key academic studies that form the theoretical foundation of this project.

2.2 Early Statistical Approaches

Neslin et al. (2006) conducted one of the earliest comprehensive reviews of churn management and identified logistic regression as the baseline model for churn prediction due to its interpretability and computational simplicity. This project confirms their finding — Logistic Regression achieved the highest accuracy (80.48%) among the three models tested.

2.3 Machine Learning Approaches

Burez and Van den Poel (2009) compared multiple machine learning models including decision trees, random forests, and neural networks on telecom churn data. While they found ensemble methods generally outperform logistic regression, this project demonstrates that on well-preprocessed datasets with moderate size (7,043 records), Logistic Regression can be competitive or superior due to reduced overfitting risk.

2.4 XGBoost and Gradient Boosting

Chen and Guestrin (2016) introduced XGBoost as a dominant algorithm for structured data. In this project, XGBoost achieved 78.57% accuracy with an AUC of 0.821, demonstrating strong performance though it was outperformed by Logistic Regression on this specific dataset.

2.5 Feature Importance Research

Coussement and Van den Poel (2008) emphasized the importance of feature engineering. This project confirms their findings — the top churn predictors identified through Random Forest feature importance are TotalCharges, MonthlyCharges, Tenure, Internet Service type (Fiber Optic), and Payment Method (Electronic check).

2.6 Business Impact of Churn Reduction

Gupta et al. (2006) found that a 5% increase in customer retention leads to profit increases of 25% to 95%. With our model achieving 80.48% accuracy and AUC of 0.842, businesses can proactively identify approximately 8 out of 10 potential churners and take retention action before they leave.

2.7 Research Gap

While extensive research exists on churn prediction algorithms, fewer studies focus on end-to-end deployment with real-time dashboards for non-technical business users. This project addresses this gap by combining predictive modeling with practical deployment recommendations.

CHAPTER 3: RESEARCH OBJECTIVES AND METHODOLOGY

3.1 Research Objectives

- To identify and analyze key factors influencing customer churn using exploratory data analysis.
- To develop and compare ML models (Logistic Regression, Random Forest, XGBoost) for churn prediction.
- To evaluate models using Accuracy, Precision, Recall, F1-Score, AUC, and Cross-Validation.
- To build an interactive dashboard visualizing churn insights for business decision-makers.
- To deploy the final model as a REST API using Flask for CRM integration.

3.2 Research Problem

Businesses in the telecom industry face significant revenue losses due to customer churn. Most organizations lack the predictive mechanisms to identify at-risk customers before they leave. This project builds an automated, accurate AI-based system that predicts churn probability (achieving 80.48% accuracy) and recommends retention strategies.

3.3 Research Design

Applied quantitative research design following the CRISP-DM methodology: Business Understanding → Data Understanding → Data Preparation → Modeling → Evaluation → Deployment.

3.4 Type of Data Used

Secondary quantitative data — structured customer records including demographics, account details, and service usage patterns from a real telecom company.

3.5 Data Collection Method

Dataset sourced from Kaggle (IBM Telco Customer Churn dataset) — 7,043 real-world telecom customer records with 21 attributes.

3.6 Sample Size

- Total Records: 7,043 customers
- Training Set: 5,634 records (80%)
- Test Set: 1,409 records (20%)

3.7 Sampling Technique

Stratified random sampling to maintain 26.5% churner vs 73.5% non-churner proportion in both training and test sets.

3.8 Data Analysis Tools

- Programming Language: Python 3.10 | Environment: Google Colab
- Data Processing: Pandas, NumPy
- Visualization: Matplotlib, Seaborn
- Machine Learning: Scikit-learn (Logistic Regression, Random Forest), XGBoost
- Deployment: Flask REST API | Model Saving: Pickle

CHAPTER 4: DATA ANALYSIS, RESULTS AND INTERPRETATION

4.1 Data Preprocessing Results (Week 1-2)

- Missing Values: 11 missing values in TotalCharges — imputed with median.
- Feature Removal: customerID column dropped (no predictive value).
- Encoding: Binary columns label encoded; InternetService, Contract, PaymentMethod one-hot encoded.
- Scaling: tenure, MonthlyCharges, TotalCharges standardized using StandardScaler.
- Final Dataset: 7,043 rows × 24 features | Train: 5,634 | Test: 1,409

4.2 EDA Key Findings (Week 3-4)

- Overall Churn Rate: 26.5% (1,869 out of 7,043 customers).
- Contract Type: Month-to-month = ~42% churn | Two-year = ~3% churn.
- Tenure: New customers (0-10 months) have highest churn risk.
- Internet Service: Fiber optic = ~33% churn vs DSL = ~19% churn.
- Monthly Charges: Churned customers pay higher average monthly charges.
- Correlation: Tenure-TotalCharges = 0.83 | Tenure-Churn = -0.35.

4.3 Model Training Results (Week 5-6)

Three ML models trained on 5,634 records and evaluated on 1,409 test records:

Model	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression ⭐	80.48%	65.71%	55.35%	60.09%	0.842
Random Forest	78.64%	62.29%	49.47%	55.14%	0.827
XGBoost	78.57%	60.91%	53.74%	57.10%	0.821

4.4 Cross Validation Results (5-Fold)

Model	Mean CV Accuracy	Std Deviation
Logistic Regression ⭐	80.43%	0.77%
Random Forest	78.89%	1.09%
XGBoost	78.21%	1.01%

Logistic Regression is the most consistent model with highest mean CV accuracy (80.43%) and lowest standard deviation (0.77%), confirming its stability across different data splits.

4.5 Top 10 Important Features (Random Forest)

Feature importance analysis identified the following key churn predictors in order of importance:

- 1. TotalCharges — Most important predictor (score ~0.19)
- 2. MonthlyCharges — Strong billing-related predictor (~0.17)
- 3. Tenure — Customer loyalty indicator (~0.15)
- 4. InternetService_Fiber optic — Service type indicator
- 5. PaymentMethod_Electronic check — Payment behavior
- 6. Contract_Two year — Contract commitment level
- 7. PaperlessBilling — Digital engagement indicator
- 8. gender — Demographic factor
- 9. OnlineSecurity — Value-added service usage
- 10. Partner — Customer relationship status

CHAPTER 5: FINDINGS AND CONCLUSION

5.1 Key Findings

Finding 1 - Logistic Regression is the Best Model (AUC: 0.842): The simpler Logistic Regression model achieved the highest accuracy (80.48%) and AUC (0.842) among all three models. This finding is significant as it demonstrates that model complexity does not always guarantee better performance. On this well-preprocessed dataset of 7,043 records, Logistic Regression's linear decision boundary proved more effective than ensemble methods, which showed signs of overfitting on the training data.

Finding 2 - Contract Type Drives Churn Most Strongly: Month-to-month contract customers churn at approximately 42% compared to only 3% for two-year contract customers — a 39-percentage-point difference. This is the most actionable business insight from the analysis. Telecom companies should design aggressive incentive programs to migrate customers from month-to-month contracts to longer-term commitments, as this single factor could dramatically reduce overall churn.

Finding 3 - Billing Variables are Top Predictors: TotalCharges and MonthlyCharges emerged as the two most important features in the Random Forest feature importance analysis. Combined with tenure as the third most important feature, this confirms that the customer's financial relationship with the company — how much they pay and how long they have been paying — is the strongest set of churn predictors. Customers who pay high monthly charges for shorter durations are at highest risk.

Finding 4 - All Models Show Strong AUC Performance: Despite differences in accuracy, all three models achieved AUC scores above 0.82, indicating strong discriminative ability. The Logistic Regression AUC of 0.842 means the model correctly ranks a randomly selected churner above a randomly selected non-churner 84.2% of the time — a commercially viable performance level for a churn prediction system.

5.2 Conclusion

This project successfully built a complete AI-based customer churn prediction system. Through data collection, preprocessing, EDA, model training, and comprehensive evaluation, the Logistic Regression model emerged as the optimal solution with 80.48% accuracy, 65.71% precision, 55.35% recall, 60.09% F1-Score, and 0.842 AUC on the Telco Customer Churn dataset.

The key business insight is clear: contract type, billing amounts, and customer tenure are the primary drivers of churn. By implementing the AI-based prediction system alongside targeted retention strategies for high-risk customer segments, telecom businesses can proactively reduce churn, protect revenue, and improve customer lifetime value significantly.

CHAPTER 6: RECOMMENDATIONS AND LIMITATIONS

6.1 Recommendations

- Deploy Logistic Regression Model: The 80.48% accuracy model (AUC: 0.842) should be deployed as a weekly churn scoring system to identify at-risk customers proactively.
- Incentivize Long-Term Contracts: Offer 15-20% discounts to migrate month-to-month customers to annual/two-year contracts — the strongest churn reduction lever.
- Focus on New Customer Onboarding: Assign dedicated success managers to customers in their first 6 months — the highest churn risk period.
- Address Fiber Optic Churn (~33%): Investigate pricing, service reliability, and competition for fiber optic customers showing elevated churn.
- Introduce Loyalty Programs: Reward long-tenure customers with exclusive benefits and service upgrades.
- Apply SMOTE Oversampling: Improve model recall (currently 55.35%) by addressing class imbalance (73.5% vs 26.5%).
- Automate CRM Integration: Trigger retention workflows automatically when churn probability exceeds 60%.
- Personalized Retention Offers: Create segment-specific offers for senior citizens, high-charge customers, and fiber optic users.
- Quarterly Model Retraining: Refresh model with new customer data every quarter to maintain accuracy.
- Hyperparameter Tuning: Apply GridSearchCV to potentially improve XGBoost and Random Forest performance.
- Expand Feature Set: Include customer service call frequency, complaint history, and competitor pricing data.
- Build Real-Time Dashboard: Deploy Streamlit dashboard with live churn predictions for business teams.

6.2 Limitations of the Study

- Single Company Dataset: Results from one telecom company may not generalize across different providers or industries.
- Static Dataset: No real-time data; model may degrade on live streaming customer behavior data.
- Class Imbalance: 73.5% vs 26.5% split reduces recall; SMOTE/class weighting not yet applied.
- Limited Features: Dataset lacks customer service calls, social media sentiment, and competitor pricing data.
- No Causal Analysis: Correlations identified but causal relationships for churn drivers not established.
- Computational Constraints: Google Colab free tier limited hyperparameter search depth.
- Deployment Scope: Full enterprise CRM integration requires additional infrastructure beyond this project scope.

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